Physics of Fluids

RESEARCH ARTICLE | JUNE 07 2023

Post-detonation fireball modeling: Validation of freeze out approximations

Special Collection: Shock Waves

Anthony A. Egeln, Jr. ; John C. Hewson ; Daniel R. Guildenbecher ; Ryan T. Marinis; Marc C. Welliver; Ryan W. Houim

Check for updates

Physics of Fluids 35, 066117 (2023)
https://doi.org/10.1063/5.0153334

View Online
Export Citation

Connected Content

A companion article has been published: Explosive behavoir: Characterizing the freeze-out assumption in post-detonation explosive reactions

Articles You May Be Interested In

Calculation of the Detonation Properties of Solid Explosives with the Kistiakowsky-Wilson Equation of State

J. Chem. Phys. (May 1956)

Noble–Abel/first-order virial equations of state for gas mixtures resulting from multiple condensed reactive materials combustion

Physics of Fluids (January 2022)

Chemistry of Detonations. I. A Simple Method for Calculating Detonation Properties of C–H–N–O Explosives

J. Chem. Phys. (January 1968)

30 September 2026 21:40:58

AIP Publishing
Physics of Fluids
Special Topics Open for Submissions
Learn More

AIP Publishing

Physics of Fluids ARTICLE pubs.aip.org/aip/pof

Post-detonation fireball modeling: Validation of freeze out approximations

Cite as: Phys. Fluids 35, 066117 (2023); doi: 10.1063/5.0153334
Submitted: 6 April 2023 · Accepted: 21 May 2023 ·
Published Online: 7 June 2023

View Online Export Citation CrossMark

Anthony A. Egeln, Jr.,[1] John C. Hewson,[2] Daniel R. Guildenbecher,[2] Ryan T. Marinis,[2] Marc C. Welliver,[2] and Ryan W. Houim[1,a)]

AFFILIATIONS

[1]Department of Mechanical and Aerospace Engineering, University of Florida, Gainesville, Florida 32611, USA
[2]Sandia National Laboratories, Alburquerque, New Mexico 87185, USA

Note: This paper is part of the special topic, Shock Waves.
[a)]**Author to whom correspondence should be addressed:** rhouim@ufl.edu

ABSTRACT

A numerical simulation study was performed to examine the post-detonation reaction processes produced by the detonation of a 12 mm diameter hemispherical pentaerythritol tetranitrate (PETN) explosive charge. The simulations used a finite rate detailed chemical reaction model consisting of 59 species and 368 reactions to capture post-detonation reaction processes including air dissociation from Mach 19+ shock waves that initially break out of the PETN charge, reactions within the detonation products during expansion, and afterburning when the detonation products mix with the shock heated air. The multi-species and thermodynamically complete Becker–Kistiakowsky–Wilson real-gas equation of state is used for the gaseous phase to allow for the mixing of reactive species. A recent simplified reactive burn model is used to propagate the detonation through the charge and allow for detailed post-detonation reaction processes. The computed blast, shock structures, and mole fractions of species within the detonation products agree well with experimental measurements. A comparison of the simulation results to equilibrium calculations indicates that the assumption of a local equilibrium is fairly accurate until the detonation products rapidly cool to temperatures in the range of 1500–1900 K by expansion waves. Below this range, the computed results show mole fractions that are nearly chemically frozen within the detonation products for a significant portion of expansion. These results are consistent with the freeze out approximation used in the blast modeling community.

Published under an exclusive license by AIP Publishing. https://doi.org/10.1063/5.0153334

30 September 2026 21:40:58

I. INTRODUCTION

The detonation of an explosive charge produces a high-Mach blast wave and fuel-rich products. Afterburning occurs when the detonation products mix and react with the shock-heated air.[1] These post-detonation reaction processes act on time scales that are orders of magnitude longer than the detonation itself,[2] can produce over half of the chemical energy released for explosives, such as trinitrotoluene (TNT),[3] and contribute heavily to quasi-static overpressure.[4]

The detonation of an uncased explosive charge can be broken into five stages.[5] Stage 1 occurs as the detonation moves through the charge, turning the solid explosive into hot, dense detonation products. Stage 2 occurs when the detonation wave reaches the charge surface, releasing a powerful shock wave that dissociates the ambient air and can raise local temperatures in excess of 12 000 K. At the same time, the detonation products expand outward, and Rayleigh–Taylor instabilities form at the interface with the shock heated air. Overexpansion of the detonation products also results in a secondary shock that travels inward with respect to the leading edge of the detonation products. Because the bulk flow is rapidly expanding, the absolute motion of this secondary shock is initially radially outward; however, after some time, the expansion slows, and in stage 3, the secondary shock begins to return toward the center of the charge. Stage 4 occurs when this shock impacts the center of the domain and reflects outward through the fireball, forming Richtmyer–Meshkov instabilities, which greatly enhance mixing. Finally, in stage 5, the detonation products mix with air and burn, forming the post-detonation fireball.

Numerical simulations of stages 4 and 5 turbulent mixing and afterburning generally require reactive computational fluid dynamics (CFD). Prior efforts to simulate these fireballs have solved the high Reynolds number Euler equation using implicit large eddy simulations (LES).[1,5–13] Several efforts also included Lagrangian simulation of discrete particle dynamics.[5,10,12] In general, LES predictions have been shown to reasonably match experimentally observed hydrodynamics and overall fireball structures.

Published under an exclusive license by AIP Publishing

Several approaches have been proposed to predict chemical energy release and reactions in the post-detonation fireball. Due to the complexity, model simplifications are inherently necessary. Several works by Kuhl *et al.*[1,6–8,12] and others[9] have focused on single-step, infinite-rate reaction mechanics. Alternatively, several simple kinetics models have been explored. For example, Fedina *et al.*[5,10] implemented the two-step reduced kinetics model of TNT detonation by Tran *et al.*[14] The authors were able to differentiate between global carbon and aluminum reaction rates using this approach.

Prior work provides limited insight into the evolving chemical composition within post-detonation fireballs due to the use of infinite rate or simplified kinetics models. The investigation of such dynamics generally requires detailed multi-species and multi-step reaction mechanisms,[15] which add significant computational complexity. This has been demonstrated by the large body of research on the gas-phase[16–21] and multiphase spray detonation modeling.[22–24] The current work is the first known attempt to combine LES modeling of explosive fireballs with such detailed gas-phase chemical kinetics modeling. The results presented here reveal new modeling opportunities and identify several ongoing challenges.

All simulations of post-detonation fireballs, including the current work, benefit from experimental validation. Historically, measurements have been mostly limited to pressure gauges and standoff imaging. As demonstrated in the prior work,[1,6,7,10] these measured quantities are primarily controlled by gas dynamics and provide limited utility to validate thermal and reaction predictions.

Recently, several laser spectroscopic techniques have been demonstrated in post-detonation environments.[25–30] These non-obtrusive diagnostics probe specific molecular transitions and are sensitive to local temperatures, making them attractive choices for model validation.[29] However, these measurements are also relatively sensitive to chemical compositions and minor species, such as OH or CN. As a result, the improved use of laser diagnostics and interpretation of results also necessitates companion simulations that predict detailed chemical compositions. Therefore, a second motivation of the current work is to advance a modeling framework that is well suited to future experimental validation efforts.

Finally, most prior LES predictions have been initialized by theory corresponding to sometime after detonation. For example, Taylor's self-similar solution[31] is commonly applied to initialize a spherical charge at the moment of shock breakout (stage 2).[7] As such, these predictions do not directly model the initial evolution of the detonation wave in the explosive charge (stage 1) and are incapable of simulating more complex, non-spherical explosive devices. Simulation of such complex systems requires the merging of explosive hydrodynamic modeling, which is the traditional domain of hydrocodes and related techniques,[32–34] with later-time LES simulation of turbulent mixing and afterburning. Toward this end, the current work presents a first demonstration of our recently proposed reactive burn model[35] to simulate the dynamics of stages 1–5 directly within a single LES framework.

As outlined above, the current work proposes several novel advancements for the simulation of explosive devices and post-detonation fireballs. The primary goals of this work are to (1) define the modeling framework, (2) demonstrate consistency with prior experiments and known detonation kinetics, and (3) discuss opportunities for new physical insights. Note that the current work employs common explosive equations of state (EOS)[36,37] and kinetics mechanisms.[38] The literature contains many alternative EOS variants[2,39] and kinetic mechanisms.[40] The current work does not attempt to determine which of these EOS and kinetic mechanisms are most predictive and instead defines a framework, which could be used to do so in the future.

The thermodynamic state produced by a detonation wave is initially described by the Chapman–Jouguet (CJ) conditions where high-temperature and pressure drive extremely high reaction rates. As a result, at CJ conditions, chemical compositions are well approximated by the local equilibrium.[39,41] As the post-detonation products cool by expansion, reaction rates continually slow and eventually become negligible. After this time, the chemical composition of the post-detonation products is approximately frozen. This phenomenon is known as the "freeze out assumption" and is commonly employed in blast modeling. Often, detonation products are assumed to be in the local chemical equilibrium until the temperature falls below a threshold value (typically in the range of 1500–1900 K[42,43]). Below this temperature, reactions are assumed to halt, and the chemical composition is unchanged. This process is rapid, and experimentally measured compositions of post-detonation products without mixing and reactions with air[42,44] tend to show good agreement with frozen compositions calculated after just a few volume expansions.[45]

In a typical explosive device, including the current simulations, chemical compositions of the post-detonation products are approximately frozen well before significant mixing and oxidation with the shock heated air in stages 4 and 5. It is therefore paramount that the simulation framework proposed here accurately captures the freeze out dynamics, to produce correct species within the fireball prior to afterburning. For this reason, the current work focuses on the validation of the predicted freeze out dynamics. This is a prerequisite to build confidence in future work where we will analyze stages 4 and 5 fireball dynamics in detail.

The remainder of this work considers hemispherical pentaerythritol tetranitrate (PETN) charges using 2D axisymmetric and 3D reactive flow simulations. PETN was selected because its detonation products contain negligible soot,[43] equation of state (EOS) parameters are widely available,[36,37] and several prior works provide experimental validation data.[42–44] The current PETN charges are also being quantified in ongoing experimental efforts.

The first half of this paper discusses the numerical and geometrical setup. In the second half, comparisons to experimental results are made and the blast dynamics are described. A method for exploring the chemical freeze out approximation is then described and presented. Several parameters and their influence on the results are considered, including the influence of length scale, which is explored through a parametric study of the PETN charge diameter.

II. GEOMETRICAL AND NUMERICAL SETUP

A. Geometrical setup

The geometrical and computational setup for the simulations performed in this study are shown in Fig. 1. A 12 mm diameter hemispherical pentaerythritol tetranitrate (PETN), $C_5H_8N_4O_{12}$, charge is detonated in air at 298 K and 1 bar. The detonation is center-initiated, and the initial density of the PETN is 1600 kg/m^3. PETN was chosen due to its low oxygen balance (−10%) and relatively transparent fireball,[46] which simplifies diagnostics in ongoing experiments. The charge size was varied by scaling all geometry length and time

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

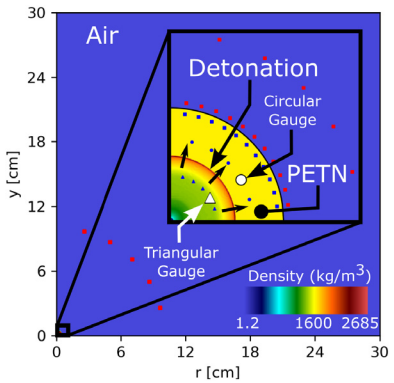


FIG. 1. The geometrical setup of the numerical simulation showing the detonation as it propagates through the charge for case 1. Symmetry boundary conditions are applied at all domain boundaries. The points represent initial positions of Lagrangian tracer gauges that track chemical composition, pressure, temperature, density, activity coefficients [see Eq. (17)], and compressibility factor.

scales by factors of 10 and 100. This results in a parametric study that encompasses charges of 12, 120, and 1200 mm diameter. The simulation setups are summarized in Table I.

The simulations were performed in 2D axisymmetric as well as 3D Cartesian coordinates, both with symmetry boundary conditions. The overall domain for the 2D simulations is $30 \times 30\,\mathrm{cm}^2$. The y axis is an axis of rotation, and the x-direction is the radial coordinate. The 3D simulations used quad symmetry with an overall domain of $30^3\,\mathrm{cm}^3$. Lagrangian tracer gauges are seeded in the domain in four regions: within the explosive charge, within the explosive charge near the interface, outside the explosive charge near the interface, and outside of the explosive charge. These tracers allow for the tracking of thermodynamic states throughout the flow.

B. Governing equations

An in-house code, HyBurn, was used to simulate the detonation of a hemispherical PETN charge and post-detonation reaction processes. The reactive Euler equations for a two-phase mixture consisting of a solid explosive (subscript h) and a reactive gas (subscript g)[35] are solved. The governing equations are

$$\frac{\partial \alpha_h}{\partial t} + \nabla \cdot (\alpha_h \mathbf{v}) - \alpha_h \nabla \cdot \mathbf{v} = -\dot{\alpha}, \tag{1}$$

$$\frac{\partial \alpha_h \rho_h}{\partial t} + \nabla \cdot (\alpha_h \rho_h \mathbf{v}) = -\dot{M}, \tag{2}$$

$$\frac{\partial \alpha_g \rho_g Y_{g,i}}{\partial t} + \nabla \cdot (\alpha_g \rho_g Y_{g,i} \mathbf{v}) = \alpha_g \dot{\omega}_i + \dot{M}_i, \tag{3}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v}) + \nabla p = 0, \tag{4}$$

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot [\mathbf{v}(\rho E + p)] = 0, \tag{5}$$

where α_h is the explosive volume fraction, α_g is the gas volume fraction, ρ is the density, $\mathbf{v}$ is the velocity vector, p is the pressure, E is the total energy, and $Y_{g,i}$ and $\dot{\omega}_i$ are the mass fraction and homogeneous mass rate of production for gas-phase species i. The terms $\dot{M}$, $\dot{\alpha}$, and $\dot{M}_i$ represent the conversion of solid explosive mass and volume to the gas phase and the production of gas-phase species i due to solid-phase reactions. By definition of α_g and α_h, we get

$$\alpha_g + \alpha_h = 1. \tag{6}$$

The total energy is

$$E = e + \frac{1}{2}\mathbf{v} \cdot \mathbf{v}, \tag{7}$$

where e is the internal energy, which includes chemical energy. As a mixture, the mass and energy, respectively, are taken to be

$$\rho = \alpha_g \rho_g + \alpha_h \rho_h, \tag{8}$$

$$\rho e = \alpha_g \rho_g e_g + \alpha_h \rho_h e_h, \tag{9}$$

where e_g and e_h are the internal energies for gas and explosive.

The Becker–Kistiakowsky–Wilson (BKW) EOS is used to model the gas-phase EOS.[36] BKW is a covolume EOS that performs well enough at explosive detonation conditions, limits to an ideal gas at low pressure, allows for mixtures of reactive species, and is thermodynamically complete and consistent. Other EOSs may perform better, such as the Jacobs–Cowperthwaite–Zwisler (JCZ) EOS;[39] however, it requires more substantial derivations and may not have parameters that are public access. The BKW EOS is

$$p = \rho R T Z, \tag{10}$$

where R is the gas constant, and T is temperature. The compressibility factor is

$$Z = 1 + x e^{\beta x}, \tag{11}$$

where

$$x = \frac{\rho \sum K_i Y_{g,i}}{(T+\theta)^{\alpha}} \tag{12}$$

and

$$K_i = \frac{\kappa \hat{K}_i}{MW_i}. \tag{13}$$

TABLE I. Simulation setups for scaled charges.

Case	Charge diameter (mm)	Dimensionality	Domain	End time (ms)
1	12	2D axisymmetric	$0.3 \times 0.3\,\mathrm{m}^2$	0.3
2	120	2D axisymmetric	$3 \times 3\,\mathrm{m}^2$	3
3	1200	2D axisymmetric	$30 \times 30\,\mathrm{m}^2$	30
4	12	3D	$0.3 \times 0.3 \times 0.3\,\mathrm{m}^3$	0.3

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

$\hat{K}_i$ is the molar covolume factor. MW_i is the species molecular weight. The BKW parameters come from the BKWS formulation and are $\alpha = 0.5$, $\theta = 6620$, $\kappa = 10.5$, and $\beta = 0.3$.[37] The values of $\hat{K}_i$ are taken from the database of Hobbs and Baer.[36] Species not found in their database were determined using the same volume estimation procedures discussed in Ref. 36 with molecular coordinates determined using the Open Babel code.[47] Covolumes found by both methods agree within a few percent.

Thermodynamic data are based on NASA9 polynomials that are valid up to 20 000 K for many species including O_2, N_2, and NO.[48] Specific heats are assumed to be constant at the upper limit of their polynomial range for species where NASA9 polynomials could not be found. Explosive blasts also produce very low temperatures (~100 K) during the expansion process. The specific heats are assumed to be constant below the lower limit of their polynomials.

A simplified barotropic EOS was used to model the solid explosive[35]

$$p - p_0 = a^2 D_{\mathrm{cj}}^2 (\rho_{\mathrm{h}} - \rho_0), \tag{14}$$

where p_0 is the reference pressure (1 bar), ρ_0 is the initial density of the explosive (1600 kg/m^3), D_{cj} is the detonation velocity (7780 m/s), and $a = 0.707$. The internal energy for PETN neglects non-ideal compression terms[49] and is given by

$$e_h = h_{f,h}^0 + Cv_h (T - T_{\mathrm{ref}}), \tag{15}$$

where $h_{f,h}^0$ and Cv_h for PETN were -1.703 MJ/kg[42] and 1000 J/kg · K,[50] respectively, and T_{ref} is 298.15 K. A pressure–temperature equilibration procedure is applied in computational cells that contain a mixture of explosive and gas by solving Eqs. (8) and (9) to find the mixture pressure and temperature.[35]

C. Reaction model

The gas-phase reaction mechanism consists of 59 species and 368 reactions.[38] This reaction mechanism was developed primarily for propellant combustion and includes many nitrogen-based species typical of energetic materials. As a result, this mechanism can capture air dissociation by the shock as it breaks out of the PETN charge, post-detonation reaction processes inside the fireball as it expands, and afterburning processes. It is important to capture all three of these aspects of the blast, as they each impact the final result of the flow. The homogeneous reaction rate is

$$\dot{\omega}_i = \sum_j \dot{\omega}_{i,j}, \tag{16}$$

where the production rate of species i due to reaction j is

$$\dot{\omega}_{i,j} = MW_i (\nu''_{i,j} - \nu'_{i,j}) \left[k_{f_j} \prod_{i=1}^{m} (C_i)^{\nu'_{i,j}} - k_{r_j} \prod_{i=1}^{m} (C_i)^{\nu''_{i,j}} \right], \tag{17}$$

where C_i, $k_{f,j}$, $k_{r,j}$, $\nu'_{i,j}$, and $\nu''_{i,j}$ are the molar concentration of species i, forward and reverse rate constants for reaction j, and stoichiometric coefficients of the reactants and products for species i in reaction j, respectively. The activity coefficients, γ_i, are used to account for real-gas effects on chemical equilibrium[51,52] and are defined by

$$\gamma_i = \exp\left(\frac{\mu_i - \mu_i^{ig}}{RuT} \right), \tag{18}$$

where Ru is the universal gas constant, and μ_i and μ_i^{ig} are the real-gas and ideal-gas chemical potentials. Ideal-gas chemical equilibrium and reaction rates can be recovered by forcing $\gamma_i = 1$. The influence of real-gas activity coefficients is discussed in Sec. IV D. The remainder of this work will focus on results for $\gamma_i = 1$.

The reverse rate coefficients for reactions, such as $C_2N_2 + OH \rightleftharpoons HOCN + CN$, exceed 10^{60} m^3 kmol^{-1} s^{-1} at low temperatures, which is unphysical. Thus, all reactions were assumed to halt below 500 K. At such lower temperatures, the reactions should physically be nearly frozen with respect to post-detonation expansion timescales. This is an *ad hoc* solution to a known problem when using detailed chemical reaction mechanisms in reactive fluid dynamic codes.[53]

The model to describe conversion of solid explosive to gas is a simplified reactive burn approach.[35] The mass exchange rate between the explosive and gas is

$$\dot{M} = \rho_{\mathrm{h}} \dot{\alpha}, \tag{19}$$

$$\dot{M}_i = \dot{M} Y_{i,\mathrm{cj}}, \tag{20}$$

where $Y_{i,\mathrm{cj}}$ is the mass fraction of species i at the Chapman–Jouguet (CJ) state predicted by the Tiger equilibrium code for PETN at 1.6 g/cc using the JCZS3 library.[39,41] These mass fractions are listed in Table II. The rate of volume change for the explosive burn in Eq. (19) is[35]

$$\dot{\alpha} = \begin{cases} \dfrac{D_{\mathrm{CJ}}}{2N_{\mathrm{dim}}} \displaystyle\sum_{i=1}^{N_{\mathrm{dim}}} \frac{1}{\Delta x_i} & \text{if } p \geq p_{\mathrm{ign}} \\ 0 & \text{otherwise,} \end{cases} \tag{21}$$

where Δx_i is the computational cell spacing in direction i, N_{dim} is the number of dimensions, and p_{ign} is an ignition threshold with a value of 2.6 GPa. The influence of p_{ign} is discussed in Houim.[35]

D. Numerical methods

The governing equations are solved using the HyBurn code developed at the University of Florida. Spatial discretization is based on a high-order Godunov approach using the fifth-order MUSCL[54] and a modified HLLC approximate Riemann solver.[35,55] The hydrodynamic terms are advanced in time using the third-order Runge–Kutta[56] with a CFL number of 0.6. The chemical source terms are integrated with the VODE stiff

TABLE II. Initial detonation products.

Species	$Y_{i,\mathrm{cj}}$	$X_{i,\mathrm{cj}}$
CO	0.128	0.133
CO_2	0.479	0.318
H_2O	0.200	0.324
N_2	0.186	0.194
H_2	1.43×10^{-3}	2.08×10^{-2}
OH	3.13×10^{-3}	5.35×10^{-3}
NH_3	1.82×10^{-3}	3.12×10^{-3}
H	6.07×10^{-5}	1.76×10^{-3}

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

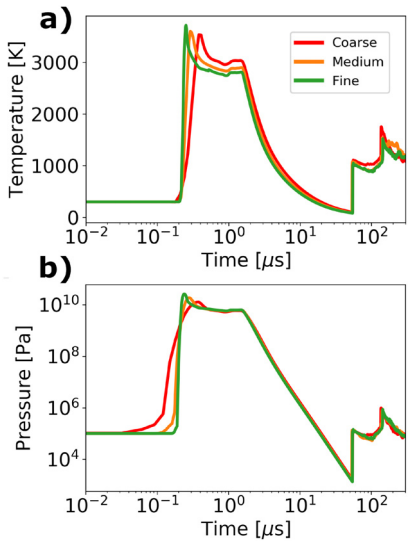


FIG. 2. (a) Temperature and (b) pressure as functions of time from the tracer gauge data of the large white triangle in Fig. 1 with minimum grid sizes of 293 (coarse), 146 (medium), and 73 μm (fine) for case 1.

ODE solver.[57] The AMReX massively parallel library is used for adaptive mesh refinement.[58] The HyBurn code has been verified against a variety of problems including premixed laminar flames, detonations, real-gas shock tube problems, and high-pressure chemical equilibrium.[16,59,60]

A grid independence study was performed with three different levels of refinement resulting in minimum grid sizes of 293 (coarse), 146 (medium), and 73 μm (fine). Figure 2 shows pressure–time and temperature–time traces from a Lagrangian tracer gauge initially located in the charge. The temperature and pressure both show convergence, and the tracer gauges experience the secondary shock at roughly the same time (~55 μs). The results indicate that the medium grid is sufficient to capture the relevant features in the simulation.

III. RESULTS

A. Validation

Figure 3 shows numerical simulation results and experimental visualization of the blast wave evolution produced by the detonation of the PETN charge. The overall progression, locations, and timing of the shock structures predicted by the simulation are in close agreement with the experimental images.

Figure 4 shows the position of the primary shock for 2D axisymmetric (case 1), 3D (case 4), and four separate experimental shots.

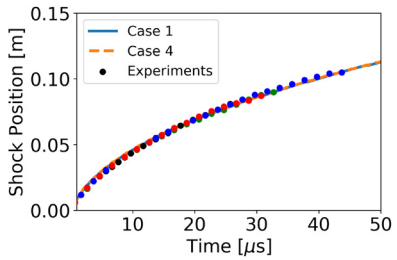


FIG. 4. The primary shock position of 2D axisymmetric (case 1), 3D (case 4), and results from four separate experimental test runs for the detonation of a 12 mm diameter hemispherical PETN charge.

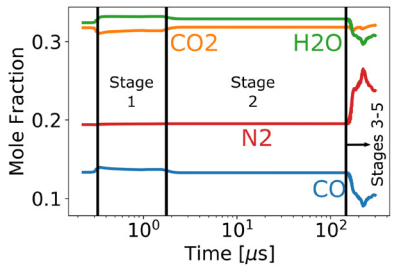


FIG. 5. Computed mole fractions of major species for the large white triangular tracer gauge shown in Fig. 1.

The simulation position was tracked by the temperature spike caused by the primary shock. The experimental positions were tracked through image processing of schlieren images, discretizing at several azimuthal angles. There is strong quantitative agreement between the numerical and experimental results.

Figure 5 shows the mole fractions of H_2O, CO_2, CO, and N_2 for the Lagrangian tracer gauge marked with the large white triangle in Fig. 1 initially located inside of the PETN charge. The mole fraction-time traces are nearly constant between 3 and 150 μs for both cases. Ornellas *et al.*[42] measured the mole fractions of the major gas species after detonating PETN in a closed bomb. A comparison of the computed mole fractions inside the detonation products between 3 and 150 μs and experimental measurements[42] are shown in Table III. There is good agreement between simulation and experimental results.

B. Overall flow and blast dynamics

Figure 6 (Multimedia view) shows the overall progression of the blast structures for case 1. Figure 7 shows azimuthally averaged position–time diagrams of the flow. As discussed in Sec. I, the flow can be

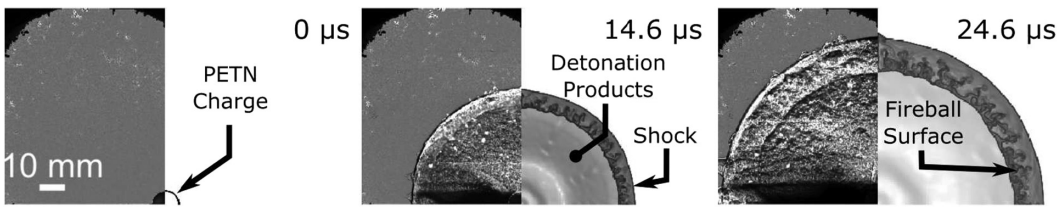


FIG. 3. A comparison between experimental (left) and numerical schlierens, $|\nabla\rho|$, (right) of the shock structure produced by the detonation of the PETN charge.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

TABLE III. Comparison of computed mole fractions inside the PETN detonation products between 3 and 150 μs to experimental measurements.

Species	Computed	Measured[42]	Difference (%)
H_2O	0.329	0.313	5.1
CO_2	0.318	0.318	0.0
CO	0.133	0.142	6.3
N_2	0.195	0.181	7.7

categorized into five stages,[5] which are labeled in Fig. 8. Stage 1 corresponds to the detonation (D1) propagating through the charge with a trailing Taylor wave (T) that expands the post-detonation flow to zero velocity near the center of the charge (see Fig. 6 at 0.52 μs.)

Stage 2 occurs when primary shock S1 of strength Mach 19+ breaks out of the charge (see Fig. 6 at 1.70 μs and Fig. 7). Shock S1 is initially strong enough to cause a high amount of air dissociation. This produces significant amounts of NO, O, and N and maximum local temperatures exceeding 12 000 K, which is consistent with

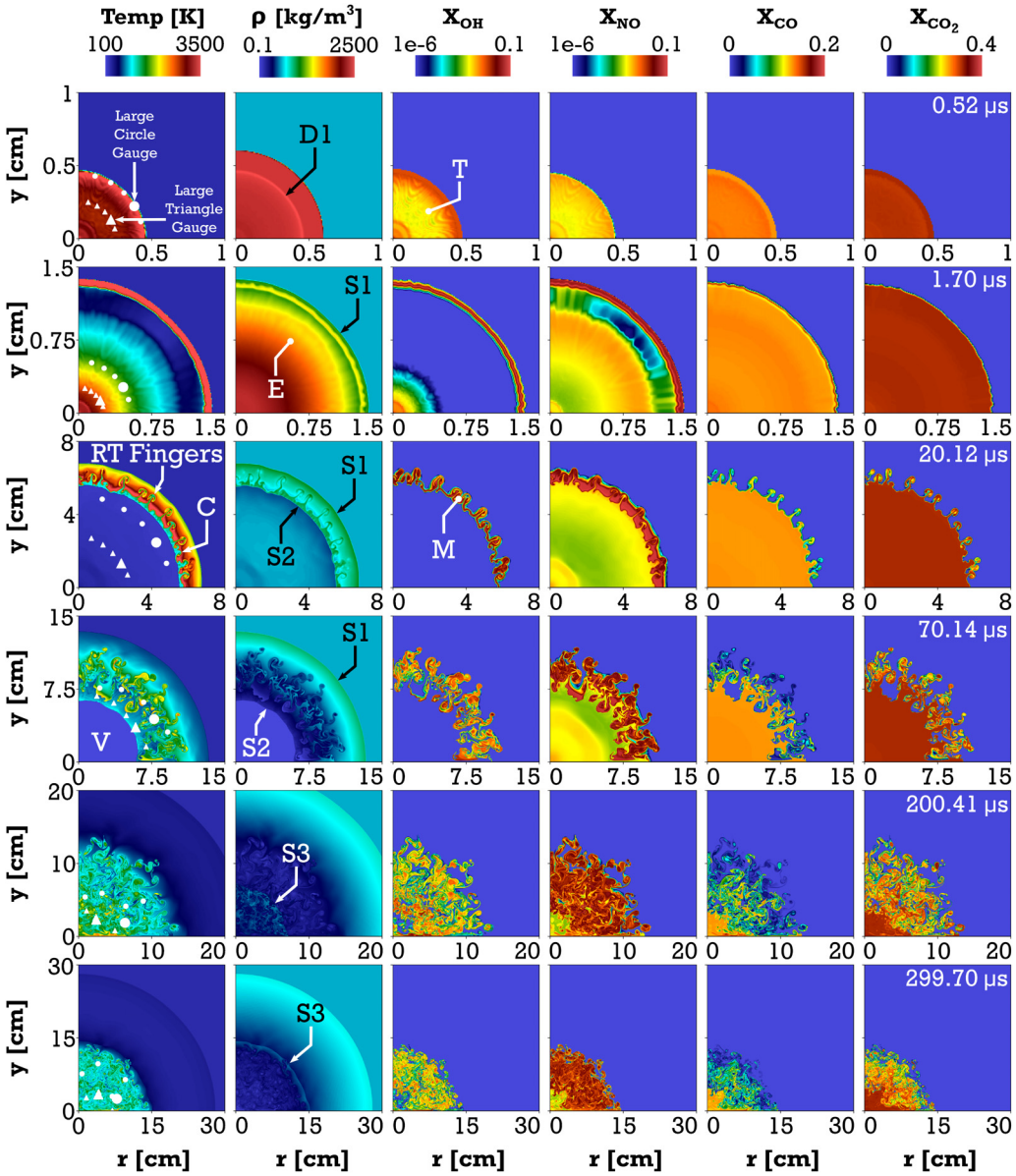


FIG. 6. Detonation of a PETN hemisphere (case 1). Left to right the columns show temperature with tracer gauge locations, density, and mole fractions of OH, NO, CO, and CO_2 at selected times. The annotations describe different aspects of the blast. D1 is the detonation. T is the Taylor wave. E is the expansion wave. C is the contact surface between the detonation products and air. M is the mixing region. V is the vacuum region. S1 is the primary shock. S2 is the secondary shock. S3 is the reshock. Multimedia available online.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

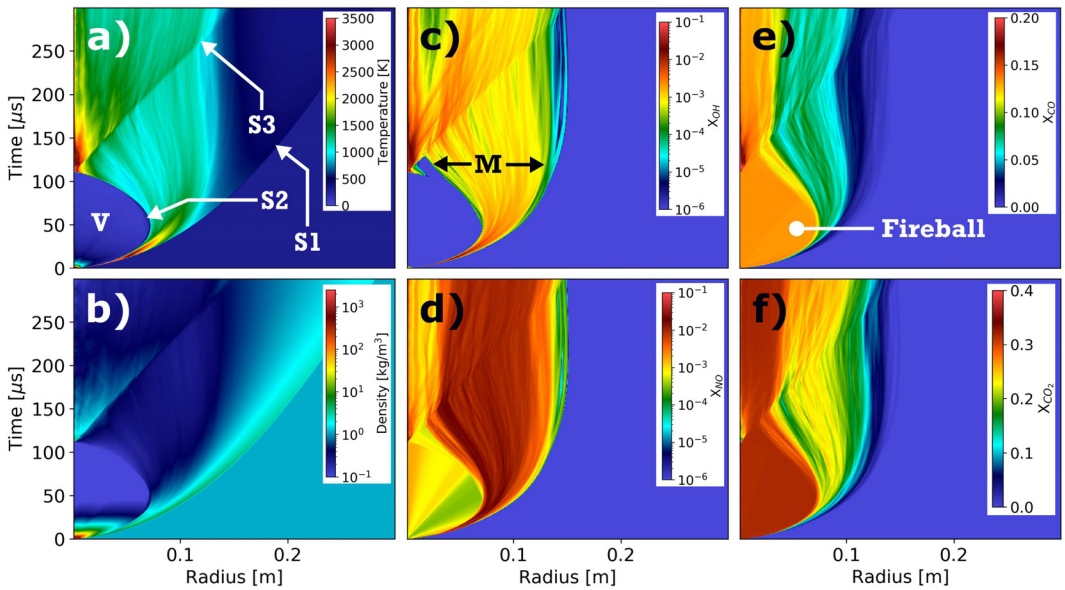


FIG. 7. Position–time diagram for case 1 showing (a) temperature, (b) density, (c) OH mole fraction, (d) NO mole fraction, (e) CO mole fraction, and (f) CO_2 mole fraction. The annotations describe different aspects of the blast. M is the mixing region. V is the vacuum region. S1 is the primary shock. S2 is the secondary shock. S3 is the reshock.

experimental measurements.[61] These species exist between the primary shock and the detonation products interface, as seen with the NO mole fraction field in Fig. 6 at 1.7 and 20.12 μs. As the blast expands and cools, many of these dissociated species are consumed or recombine. One exception is NO, which maintains high quantities throughout the mixing and afterburning processes. While the primary shock S1 propagates outward, an expansion wave (E) propagates from the outer edge of the charge through the detonation products toward the center at 1.70 μs.

Expansion wave E eventually over-expands the flow and produces secondary shock S2 inside the detonation products (see Fig. 6 at 20.12 μs and Fig. 7). Shock S2 is inward-running and is trying to propagate toward the center of the charge. However, the highly supersonic expanding flow pushes S2 radially outward forming the vacuum region V inside the fireball (see Fig. 6 at 70.14 μs and Fig. 7). Rayleigh–Taylor (RT) instabilities form on the contact surface (C) between the fuel-rich detonation products and shock-heated air. The resulting RT mixing-induced afterburning reactions form significant amounts of OH. The continued mixing of this region forms the reactive mixing layer M (see Fig. 6 at 20.12 μs and Fig. 7). High quantities of NO can be seen in region M in Fig. 7(d).

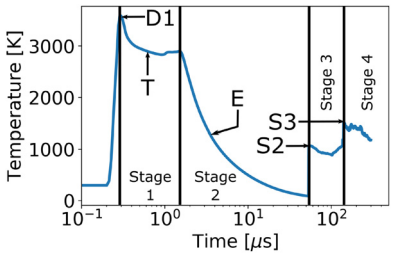


FIG. 8. Temperature-time trace for the large white triangular tracer gauge highlighting the different stages and flow features discussed in Sec. III B.

Stage 3 is the "negative phase" of the blast that begins when the expanding supersonic flow decelerates enough for secondary shock S2 to return toward the center of the charge. The flow behind the return shock S2 at this stage reverses direction and moves radially inward (see Fig. 6 at 70.14 μs). This inward motion stretches the RT "fingers" (visualized by X_{OH}) and causes them to grow. This process helps entrain air into the mixing layer,[5] increasing its size and stretching the RT instabilities. The growth of the mixing layer can be observed in Fig. 7(c).

Stage 4 is the "reshock" phase of the blast that starts when shock S2 impacts the center of the domain and reflects to produce the outward propagating shock S3 (see Fig. 6 at 200.41 and 299.70 μs). The interaction of shock S3 with the elongated RT fingers introduces Richtmyer–Meshkov (RM) instabilities and a substantial increase in mixing. The temperatures in this mixing region are ~2000 K, which is consistent with other studies.[62] As shown in Fig. 6, at early times the detonation products CO and CO_2 have similar behaviors. However, at 200.41 and 299.70 μs, the more complete combustion caused by the enhanced mixing of the RM instabilities increases the amount of CO_2. This can also be seen in Fig. 7(f) after shock S3 moves back through the mixing region M. Stage 5 (see Fig. 6 at 299.70 μs) occurs in the later stages of the post-detonation flow as fuel-rich detonation products mix with the shock-heated air and continue to burn.[5]

C. Lagrangian tracer gauges

Figures 5 and 9 show major and minor species mole fractions, respectively, recorded with the large white triangular Lagrangian tracer gauge indicated in Figs. 1 and 6. Most of the species have nearly constant mole fractions between 3 μs (shortly after the arrival of expansion wave E) and ~150 μs (shortly after shock S3 passes through the

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

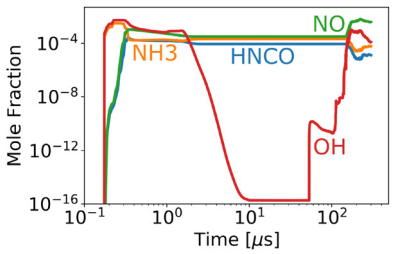


FIG. 9. Computed mole fractions of select minor species for the large white triangular tracer gauge shown in Figs. 1 and 6.

gauge). Hydroxyl (OH) is a notable exception, which continues depleting to near-zero values until ∼10 μs and responds rapidly to return shock S2. After 150 μs, the mole fractions show significant variation due to mixing with air and afterburning.

Figure 10(a) shows the mole fraction of CO and temperature for four of the triangular tracer gauges shown in Figs. 1 and 6. These gauges all start at the same radial distance from the origin, but at different azimuths. The temperature and CO mole fraction profiles are nearly identical for all gauges until ∼50 μs. Return shock S2 impacts the gauges at slightly different times due to local flow variations. This produces perturbations on return shock S2 that cause it to impact the tracer gauges at slightly different times. The differences between the gauges are more significant at later times when mixing with air and afterburning become dominant. Some of the gauges show the onset of mixing as early as 90 μs and others as late as 150 μs. The positions of the triangular tracer gauges in Fig. 6 at 70.14 μs are still all at roughly the same radial position. Some gauges begin mixing more quickly with air, while others are further away and begin mixing later. These variations are caused by the position of the gauges relative to the RT fingers.

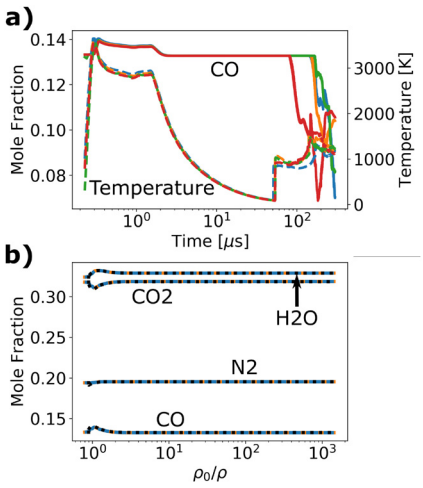


FIG. 10. Results showing (a) X_{CO} and temperature vs time for several triangular gauges shown in Figs. 1 and 6 at the same radial position and (b) mole fractions of major species as a function of ρ_0/ρ for several gauges in stages 1 and 2 where orange is the white triangular tracer gauge in Fig. 1, blue is the white circular gauge in Fig. 1, and black is from the 3D simulation.

Figure 10(b) shows the mole fraction of several particle gauges at different radial lines indicated by the large white triangle and large white circle in Fig. 1 as a function of expansion ratio (ρ_0/ρ) in stages 1 and 2. These gauges are at different radial positions and, as a result, experience different time scales. Nevertheless, the mole fractions are nearly identical between the different gauges when plotted as a function of ρ_0/ρ. The results from the 3D simulation (case 4) also show nearly identical results in Fig. 10(b) compared to the 2D simulation (case 1). This indicates that the post-detonation reactions processes are largely gas dynamic and realistic turbulence is relatively unimportant until mixing becomes important in stages 3–5.

IV. DISCUSSION

A. Assessment of the freeze out approximation

The Lagrangian tracer results shown in Figs. 5, 9, and 10 indicate nearly constant mole fraction traces during stage 2. This result is consistent with the freeze out approximation used by the blast modeling community.[1,42] The freeze out approximation assumes chemical equilibrium in the fireball as the detonation products isentropically expand until the temperature reaches a threshold value (typically 1500–1900 K[42,63]). The composition of the detonation products is typically assumed to be constant (frozen) at temperatures below this threshold value.

The accuracy of the freeze out approximation can be assessed by comparing the computational results to equilibrium calculations along the CJ expansion isentrope using Tiger with the JCZS library.[39,41] Figure 11 shows major species mole fractions and temperature as a function of expansion ratio (ρ_0/ρ) based on simulation results and Tiger equilibrium calculations. The simulation results and equilibrium calculations agree well until ρ_0/ρ exceeds 4 to 5. The equilibrium approximation becomes increasing poor as the expansion ratio increases further. For example, the equilibrium calculations predict trace amounts of CO at large expansion ratios, while the simulation results and measurements[42] show much higher mole fractions, as shown in Table III. The equilibrium assumption becomes poor when ρ_0/ρ is between 4 and 5, which corresponds to fireball temperatures between 1500 and 1900 K. This is consistent with typical freeze out temperatures used in practice.[42]

B. Influence of finite rate reactions

Figure 12(a) shows the chemical, τ_{chem}, and gas-dynamic, τ_{gas}, time scales during stages 1 and 2. The chemical time scales, τ_{chem}, are defined by solving

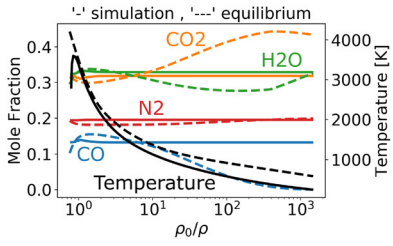


FIG. 11. Temperature and mole fractions of CO_2 (orange), H_2O (green), N_2 (red), and CO (blue) as a function of expansion ratio for the large white triangular tracer gauge in Figs. 1 and 6. The solid and dashed lines represent simulation results and Tiger[39,41] equilibrium calculations, respectively.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

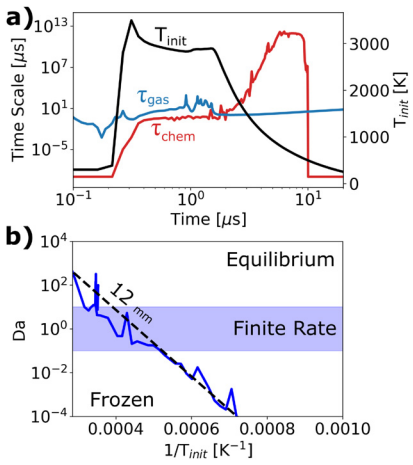


FIG. 12. Results for the large white triangular tracer gauge showing (a) temperature, τ_{chem}, and τ_{gas}, and (b) *Da* data (solid line) and the Arrhenius fit (dashed line) for the simulation of the 12 mm diameter PETN charge (case 1).

$$T(\tau_{chem}) = 0.5(T_0 + T_{eq}). \tag{22}$$

The equilibrium condition T_{eq} is approximated by integrating a constant volume reactor for 10^6 s with initial conditions, T_0, based on the tracer gauge results. The gas-dynamic time scales are defined by

$$\tau_{gas} = \frac{\rho}{|d\rho/dt|}, \tag{23}$$

where $d\rho/dt$ is calculated by finite differencing the tracer gauge results.

The chemical time scales are initially very rapid (<0.1 μs) in stage 1 when pressure and temperature are both high after the initial detonation. The chemical time scales grow in excess of 10^{13} μs in stage 2 due to gas-dynamic cooling from the isentropic expansion wave lowering the pressure and temperature that reduces the reaction rates. Gas-dynamic timescales are ~10 times slower than reaction timescales in stage 1, which indicates that the chemical composition is in near chemical equilibrium in the Taylor wave. However, τ_{chem} becomes many orders of magnitude larger than τ_{gas} in stage 2 after 4 μs. Thus, the chemical kinetics in stage 2 transitions from equilibrium near the start of stage 2, through a period where finite-rate reaction effects are important, to chemically frozen at later times.

A gas-dynamic Damköhler number can be defined by $Da = \tau_{gas}/\tau_{chem}$ to distinguish when chemical kinetics processes are in equilibrium, finite-rate, or frozen. Chemistry is frozen if $Da < Da_c \ll 1$, where $Da_c = 0.1$ is a critical Damköhler number. The computed *Da* is shown in Fig. 12(b) and is well described by an Arrhenius model,

$$Da = Ae^{-T_a/T}. \tag{24}$$

Fitting Eq. (24) to results from case 1 gives $A = 9.0 \times 10^6$ and $T_a = 35,045$ K. The freeze out temperature, T_{freeze}, can be found by solving Eq. (24) for the temperature where $Da = Da_c$, which produces

$$T_{freeze} = T_a \ln\left(\frac{A}{Da_c}\right)^{-1}. \tag{25}$$

Applying the fitted A and T_a for case 1 gives $T_{freeze} = 1913$ K, which is higher than values typically used.[42,45] Nevertheless, the equilibrium mass fractions at 1913 K, shown in Fig. 11, are close to simulation results and Ornellas *et al.*[42]

C. Influence of charge length scales

Larger PETN charge diameters of 120 (case 2) and 1200 mm (case 3) were also simulated. Arrhenius plots of *Da* for the larger charges are shown in Fig. 13, and the fitting parameters are listed in Table IV. The pre-factor, A, increases by a factor of 40 for each larger charge diameter, while T_a increases by roughly 2000 K for each larger charge diameter. The results show a consistent decline in T_{freeze} with increasing charge diameter. This is because the gas-dynamic time roughly scales in proportion to the charge diameter. The increased duration of the detonation products in the finite-rate kinetics regime brings the system closer to equilibrium before Da_c drops below 0.1.

The results from case 2 and 3 can be compared to scaled results from case 1. Gas-dynamic processes are scale invariant with time and space, while chemical reaction processes are a function of the local state; thus, gas-dynamic variables for different sized charges should be nearly identical if time and length are scaled together and finite-rate chemical kinetics are not dominant. Therefore, to first-order, the impact of freeze out on larger charges can be approximated by multiplying τ_{gas} by a scale factor, $\eta = d_{PETN}/12$ mm, where d_{PETN} is the diameter of the PETN charge. Using the definition of *Da* and applying the scale-factor, η, we find

$$Da_d = \frac{\tau_{gas,d}}{\tau_{chem}} = \eta\frac{\tau_{gas,12\,mm}}{\tau_{chem}} = \eta Da_{12\,mm}, \tag{26}$$

where Da_d and $Da_{12\,mm}$ are the Damkhöler number for a charge of diameter d and 12 mm, respectively. Substituting Eq. (24) for $\frac{\tau_{gas,12\,mm}}{\tau_{chem}}$ gives

$$Da_d = \eta A_{12\,mm}\exp\left(\frac{-T_{a,12\,mm}}{T}\right). \tag{27}$$

Solving Eq. (27) for the temperature where $Da = Da_c$ gives a scaling prediction of the freeze out temperature for different PETN charge diameters,

$$T_{freeze,predict} = T_{a,12\,mm}\ln\left(\frac{A_{12\,mm}}{Da_c}\frac{d_{PETN}}{12\,mm}\right)^{-1}. \tag{28}$$

The results of evaluating Eq. (28) with $Da_c = 0.1$ are shown in Table V as well as the freeze out temperatures of the larger-diameter

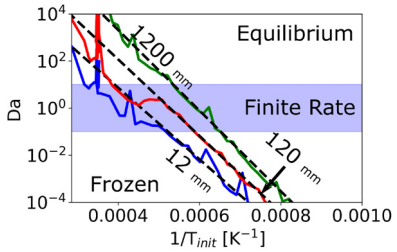


FIG. 13. Results for the large white triangular tracer gauge showing *Da* data (solid lines) and associated fits (dashed lines) for simulations of 12, 120, and 1200 mm diameter PETN charges.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

TABLE IV. Arrhenius parameters and T_{freeze} for 12, 120, and 1200 mm diameter PETN charges.

Case	Diameter (mm)	A	T_a (K)	T_{freeze} (K)
1	12	9.02×10^6	35 045	1913
2	120	3.67×10^8	37 395	1698
3	1200	1.46×10^{10}	39 410	1533

TABLE V. Predicted and simulated freeze out temperatures for 12, 120, and 1200 mm diameter PETN charges.

Case	Diameter (mm)	$T_{freeze,sim}$ (K)	$T_{freeze,predict}$ [Eq. (28)]	(K) % error
1	12	1913	1913	...
2	120	1698	1700	0.09
3	1200	1533	1529	0.29

PETN charges. The Arrhenius fits are shown in Fig. 14 where the black dashed lines are from Eq. (28). The results predict the freeze out temperatures of the larger simulations well, with less than 0.3% error. This shows that the results from simulations using smaller hemispherical charges can be used to reliably predict the freeze out temperature of much larger hemispherical charges through a length scale factor, η.

The freeze out temperature is a function of d_{PETN} due to the increased duration in the finite-rate kinetics regime with increasing charge size. This influences the freeze out temperature, but also the chemical composition and activation energy, as shown in Table VI. As the charge size increases, the mole fraction of CO_2 slightly increases and the mole fractions of H_2O and CO slightly decrease. The variation of the freeze out composition is modest for typical PETN charge diameters because the equilibrium mole fractions in Fig. 11 do not vary significantly in the range of T_{freeze}.

Figure 15 shows a comparison between the 12-mm diameter PETN charge for 2D axisymmetric (case 1) and 3D (case 4) simulations. Both the *Da* data and Arrhenius plots follow similar trends. As stated above, this analysis is performed on data from early simulation times, where the flow is primarily a non-turbulent expanding gas-dynamic flow. The thermodynamic states from the tracer gauge data are almost identical for the 2D and 3D cases, as shown by the black dotted lines in Fig. 10(b). In addition, mixing effects are not as relevant at these early times. Therefore, the freeze out temperature does not vary substantially between 2D and 3D simulations.

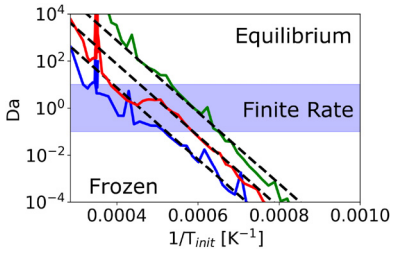


FIG. 14. Results for the large white triangular tracer gauge showing comparisons between *Da* data (solid lines) and predictions from Eq. (28) (black lines).

TABLE VI. Comparison of computed mole fractions inside the PETN detonation products for 12 (case 1), 120 (case 2), and 1200 (case 3) mm diameter charges.

Species	Case 1	Case 2	Case 3
H_2O	0.329	0.326	0.322
CO_2	0.318	0.322	0.326
CO	0.133	0.129	0.126
N_2	0.195	0.195	0.196

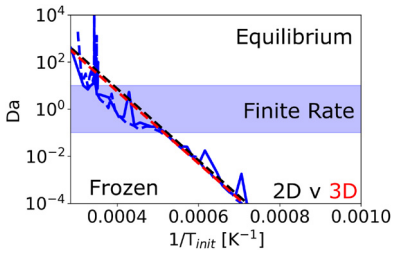


FIG. 15. Results for the large white triangular tracer gauge showing comparisons between 12 mm diameter PETN charges in 2D axisymmetric (case 1) and 3D (case 4) where the solid blue lines are 2D, the dashed blue lines are 3D, the dashed black line is the 2D Arrhenius fit, and the dashed red line is the 3D Arrhenius fit.

D. Influence of real-gas activity coefficients, γ_i

Figure 16 shows the mole fractions of H_2O, CO_2, CO, and N_2 for the Lagrangian tracer gauge marked with the large white triangle in Figs. 1 and 6 initially located inside of the PETN charge. The results for both ideal- and real-gas activity coefficients are shown. A comparison of the computed mole fractions inside the detonation products between 3 and 150 μs and experimental measurements[42] are shown in Table VII. Computations using real-gas γ_i have closer agreement for H_2O, while the results using ideal-gas γ_i have closer agreement for both CO_2 and CO.

Figure 17(a) shows the real-gas activity coefficients for CO_2, CO, H_2O, and N_2 for the large white triangular tracer gauge in Figs. 1 and 6. The values of γ_i exceed 10^7 for some major species near the detonation wave where $p > 20$ GPa. The large γ_i produces reaction rates [see Eq. (17)] that are in the equilibrium regime ($Da \gg 1$) during stage 1 with $\tau_{chem} \sim 10^{-14}$ s.

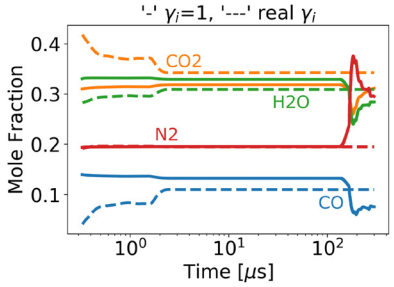


FIG. 16. Computed mole fractions of selected species as a function of time for the large white triangular tracer gauge shown in Figs. 1 and 6. Ideal- and real-gas activity coefficients are shown with the solid and dashed lines, respectively.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

TABLE VII. Comparison of computed mole fractions inside the PETN detonation products using real-gas and ideal ($\gamma_i = 1$) activity coefficients to experimental measurements.

Species	Real γ_i	$\gamma_i = 1$	Measured[42]
H_2O	0.309	0.329	0.313
CO_2	0.342	0.318	0.318
CO	0.110	0.133	0.142
N_2	0.195	0.195	0.181

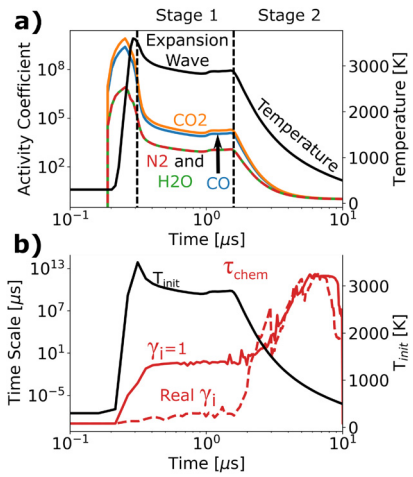


FIG. 17. The (a) activity coefficients of selected species for the large white triangular tracer gauge shown in Fig. 1 and (b) τ_{chem} calculated from simulations with ideal-gas (solid) or real-gas activity coefficients (dashed).

The decrease in density during stage 2 reduces γ_i from 10^5 to $\mathcal{O}(10)$. This substantially decreases the reaction rate and causes τ_{chem} for the ideal- and real-gas γ_i to approach each other, as shown in Fig. 17(b). The value of γ_i is 10 or less at T_{freeze}, which causes the mole fractions to be slightly different using ideal- and real-gas γ_i as shown in Fig. 16. However, these differences are modest, and either assumption produces results that agree with measurements at freeze out as shown in Table VII. As the blast continues to expand out, γ_i tends to 1. Assuming $\gamma_i = 1$ is recommended for simulations due to decreased stiffness of the chemical source terms at early times.

V. SUMMARY AND CONCLUSIONS

A modeling framework was presented that consists of a series of 2D axisymmetric and 3D numerical simulations of post-detonation reaction processes produced by the detonation of a PETN charge using a detailed chemical reaction model with 59 species and 368 reactions and the BKW real-gas EOS. The computed results show the formation of shock and blast structures that are in agreement with the literature.

The model presented here seamlessly couples the detonation of an explosive charge to afterburning processes with finite-rate detailed chemical kinetics. This method reproduced the shock and blast structures observed in experiments. In addition, it can correctly capture dynamic chemical processes, such as the freeze out condition from first principles. This provides validation for the simulation approach and allows it to be extended to future work.

The mole fractions recorded by Lagrangian tracer gauges, initially placed in the PETN charge, show a broad region with non-equilibrium chemically frozen values that agree well with experimental measurements. The gas composition is frozen in the fireball due to rapid cooling from the isentropically expanding detonation products, which decreases the reaction rates and ultimately the Damköhler number below 0.1. A comparison of simulation results with equilibrium mole fractions shows that the assumption of local chemical equilibrium is fairly accurate until the temperature falls in the range of 1900–1500 K. These freeze out temperatures are consistent with the chemically frozen regime ($Da < 0.1$) for a range of reasonably sized PETN charge diameters. The results support the freeze out assumption commonly used in blast modeling. The results show that smaller charges can be used to accurately predict the freeze out temperature of larger charges, which can provide a reduction in the computation time. The results show that assuming activity coefficients equal to unity ($\gamma_i = 1$) produces freeze out mole fractions that are in close agreement with measurements and applying real-gas activity coefficients is not important to detailed simulations for stages 2–5.

Ongoing work will extend this framework to other aspects of post-detonation flow. This includes performing three-dimensional numerical simulations to understand the role of shock-induced mixing and turbulence and finite-rate chemical reactions on late-term afterburning processes in stages 4 and 5. An analysis of several other explosive materials will be performed to explore the influence of energetics with different oxygen balances. The inclusion and influence of soot particles is under exploration. Reductions to the chemical mechanism are also ongoing.

SUPPLEMENTARY MATERIAL

The chemical mechanism used for this work has been included for reference. It consists of 59 species and 368 reactions and includes thermodynamic data based on NASA9 polynomials that are valid to 20 000 K for many species.[48] The mechanism was primarily developed for propellant combustion,[38] but includes many species relevant to explosive detonations.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the support of the Laboratory Directed Research and Development (LDRD) Program at Sandia National Laboratories. Sandia National Laboratories is a multi-mission laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC., a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under Contract No. DE-NA0003525. The views expressed in the article do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

Author Contributions

Anthony August Egeln Jr.,: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **John Hewson:** Conceptualization (equal); Supervision (equal); Validation (equal). **Daniel R. Guildenbecher:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Ryan T. Marinis:** Data curation (supporting); Validation (equal). **Marc C. Welliver:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal). **Ryan Houim:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1]A. L. Kuhl, J. B. Bell, V. E. Beckner, K. Balakrishnan, and A. Aspden, "Spherical combustion clouds in explosions," Shock Waves **23**, 233–249 (2013).
[2]L. Donahue, F. Zhang, and R. Ripley, "Numerical models for afterburning of TNT detonation products in air," Shock Waves **23**, 559–573 (2013).
[3]A. Kuhl, "Mixing in explosions," Report No. UCRL-JC-115690 (Lawrence Livermore National Laboratory, El Segundo, CA, 1993).
[4]A. L. Kuhl, "Spherical mixing layers in explosions," in *Dynamics of Exothermicity* (Gordon and Breach Publishers, 1996), Vol. 2, pp. 291–323.
[5]F. F. Grinstein, *Coarse Grained Simulation and Turbulent Mixing* (Cambridge University Press, 2016).
[6]A. Kuhl, J. Bell, and V. Beckner, "Heterogeneous continuum model of aluminum particle combustion in explosions," Combust., Explos., Shock Waves **46**, 433–448 (2010).
[7]A. Kuhl, J. Bell, V. Beckner, and H. Reichenbach, "Gasdynamic model of turbulent combustion in TNT explosions," Proc. Combust. Inst. **33**, 2177–2185 (2011).
[8]A. Kuhl and K. Balakrishnan, "Gasdynamic model of dilute two-phase combustion fields," Combust., Explos., Shock Waves **48**, 544–560 (2012).
[9]D. Grote, A. L. Kuhl, J. B. Bell, and V. E. Beckner, "Modeling optical emissions from HE fireballs," in *Proceedings of the 25th International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS)* (2015).
[10]E. Fedina and C. Fureby, "Analysis of heat-release during TNT/aluminum afterburning by means of numerical simulations," Proc. Combust. Inst. **36**, 2841–2848 (2017).
[11]S. Courtiaud, N. Lecysyn, G. Damamme, T. Poinsot, and L. Selle, "Analysis of mixing in high-explosive fireballs using small-scale pressurised spheres," Shock Waves **29**, 339–353 (2019).
[12]A. Kuhl, D. Grote, and J. Bell, "Scaling turbulent combustion fields in explosions," Appl. Sci. **10**, 8577 (2020).
[13]J. W. Posey, B. Roque, S. Guhathakurta, and R. W. Houim, "Mechanisms of prompt and delayed ignition and combustion of explosively dispersed aluminum powder," Phys. Fluids **33**, 113308 (2021).
[14]T. Tran, R. Simpson, J. Maienschein, and C. Tarver, "Thermal decomposition of trinitrotoluene (TNT) with a new one-dimensional time to explosion (ODTX) apparatus," Report No. UCRL-JC-141597 (Lawrence Livermore National Laboratory, Livermore, CA, 2001).
[15]E. Fedina, C. Fureby, G. Bulat, and W. Meier, "Assessment of finite rate chemistry large eddy simulation combustion models," Flow, Turbul. Combust. **99**, 385–409 (2017).
[16]R. W. Houim and B. D. Taylor, "Detonation initiation from shock and material interface interactions in hydrogen-air mixtures," Proc. Combust. Inst. **37**, 3513–3520 (2019).
[17]J. Fujii, Y. Kumazawa, A. Matsuo, S. Nakagami, K. Matsuoka, and J. Kasahara, "Numerical investigation on detonation velocity in rotating detonation engine chamber," Proc. Combust. Inst. **36**, 2665–2672 (2017).
[18]N. Tsuboi, S. Eto, A. K. Hayashi, and T. Kojima, "Front cellular structure and thrust performance on hydrogen–oxygen rotating detonation engine," J. Propul. Power **33**, 100–111 (2017).
[19]C. Yan, H. D. Ng, and X. Mi, "A numerical study on the influence of increased instability of quasi-detonation on the critical tube diameter phenomenon," Proc. Combust. Inst. (published online).
[20]P. Honhar, C. R. Kaplan, R. W. Houim, and E. S. Oran, "Role of reactivity gradients in the survival, decay and reignition of methane-air detonations in large channels," Combust. Flame **222**, 152–169 (2020).
[21]B. Maxwell and J. Melguizo-Gavilanes, "Origins of instabilities in turbulent mixing layers behind detonation propagation into reactive–inert gas interfaces," Phys. Fluids **34**, 106107 (2022).
[22]F. Wang and C. Weng, "Numerical research on two-phase kerosene/air rotating detonation engines," Acta Astronaut. **192**, 199–209 (2022).
[23]N. Jourdaine, N. Tsuboi, and A. K. Hayashi, "Investigation of liquid *n*-heptane/air spray detonation with an Eulerian-Eulerian model," Combust. Flame **244**, 112278 (2022).
[24]Q. Meng, M. Zhao, Y. Xu, L. Zhang, and H. Zhang, "Structure and dynamics of spray detonation in *n*-heptane droplet/vapor/air mixtures," Combust. Flame **249**, 112603 (2023).
[25]M. Soo and N. Glumac, "Ultraviolet absorption spectroscopy in optically dense fireballs using broadband second-harmonic generation of a pulsed modeless dye laser," Appl. Spectrosc. **68**, 517–524 (2014).
[26]C. Murzyn, A. Sims, H. Krier, and N. Glumac, "High speed temperature, pressure, and water vapor concentration measurement in explosive fireballs using tunable diode laser absorption spectroscopy," Opt. Lasers Eng. **110**, 186–192 (2018).
[27]M. Soo, C. Murzyn, A. Sims, J. Cerow, N. Glumac, J. Ott, M. De Magistris, N. Sinha, and J. Lightstone, "Measurement of temperature and water vapor concentration using laser absorption spectroscopy in kilogram-scale explosive fireballs," AIP Conf. Proc. **2272**, 060034 (2020).
[28]D. R. Richardson, S. P. Kearney, and D. R. Guildenbecher, "Post-detonation fireball thermometry via femtosecond-picosecond coherent anti-Stokes Raman scattering (CARS)," Proc. Combust. Inst. **38**, 1657–1664 (2021).
[29]G. C. Mathews, M. Gomez, C. J. Schwartz, A. A. Egeln, Jr., R. W. Houim, S. F. Son, M. Arienti, A. D. Thompson, M. Welliver, D. R. Guildenbecher *et al.*, "Experimental and synthetic laser-absorption-spectroscopy measurements of temperature, pressure, and CO at 1 MHz for evaluation of post-detonation fireball models," Proc. Combust. Inst. (published online).
[30]C. J. Schwartz, J. W. Stiborek, A. Butler, D. Chen, D. R. Guildenbecher, M. Welliver, N. Glumac, and C. S. Goldenstein, "Near-MHz temperature and H_2O measurements in post-detonation fireballs of 25 g hemispherical explosives using scanned-wavelength-modulation spectroscopy," Appl. Opt. **62**, 1598–1609 (2023).
[31]G. I. Taylor, "The dynamics of the combustion products behind plane and spherical detonation fronts in explosives," Proc. R. Soc. London Ser. A **200**, 235–247 (1950).
[32]J. M. McGlaun, S. Thompson, and M. Elrick, "CTH: A three-dimensional shock wave physics code," Int. J. Impact Eng. **10**, 351–360 (1990).
[33]E. Hertel, R. Bell, M. Elrick, A. Farnsworth, G. Kerley, J. McGlaun, S. Petney, S. Silling, P. Taylor, and L. Yarrington, "CTH: A software family for multi-dimensional shock physics analysis," in *Shock Waves@ Marseille I: Hypersonics, Shock Tube & Shock Tunnel Flow* (Springer, 1995), pp. 377–382.
[34]Z. S. Tabatabaei and J. S. Volz, "A comparison between three different blast methods in LS-DYNA: LBE, MM-ALE, coupling of LBE and MM-ALE," in *12th International LS-DYNA Users Conference* (2012), pp. 1–10.
[35]R. Houim, "A simplified reactive burn model for simulating explosive effects—I: Single-phase blasts," Shock Waves **31**, 851–875 (2021).
[36]M. Hobbs and M. Baer, "Nonideal thermoequilibrium calculations using a large product species data base," Shock Waves **2**, 177–187 (1992).

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing

[37]M. Hobbs and M. Baer, "Calibrating the BKW-EOS with a large product species data base and measured C-J properties," in Tenth Symposium (International) on Detonation, Boston, MA (1993).
[38]C. B. Conner and W. R. Anderson, "Modeling the combustion of JA2 and solid propellants of similar composition," Proc. Combust. Inst. **32**, 2131–2137 (2009).
[39]M. L. Hobbs, R. G. Schmitt, and H. K. Moffat, "JCZS3—An improved database for EOS calculations," Report No. SAND2018-6389C (Sandia National Laboratory, Albuquerque, NM, 2018).
[40]C. K. Westbrook and F. L. Dryer, "Chemical kinetic modeling of hydrocarbon combustion," Prog. Energy Combust. Sci. **10**, 1–57 (1984).
[41]M. Cowperthwaite and W. Zwisler, "TIGER computer program documentation," Report No. ADA002791 (Naval Ordnance Laboratory, Corona CA, 1974).
[42]D. L. Ornellas, J. H. Carpenter, and S. R. Gunn, "Detonation calorimeter and results obtained with pentaerythritol tetranitrate (PETN)," Rev. Sci. Instrum. **37**, 907–912 (1966).
[43]F. H. Ree, "A statistical mechanical theory of chemically reacting multiphase mixtures: Application to the detonation properties of PETN," J. Chem. Phys. **81**, 1251–1263 (1984).
[44]D. Ornellas, "Calorimetric determinations of the heat and products of detonation for explosives: October 1961 to April 1982," Report No. UCRL-52821 (California University Berkeley Lawrence Livermore Laboratory, 1982).
[45]S. Bastea, "Chemical equilibrium and carbon kinetics in explosives," in *Proceedings of the 15th International Detonation Symposium* (Office of Naval Research, San Francisco, CA, 2014), p. 896.
[46]S. Thiboutot, P. Brousseau, and G. Ampleman, "Deposition of PETN following the detonation of seismoplast plastic explosive," Propellants, Explos., Pyrotech. **40**, 329–332 (2015).
[47]N. M. O'Boyle, M. Banck, C. A. James, C. Morley, T. Vandermeersch, and G. R. Hutchison, "Open Babel: An open chemical toolbox," J. Cheminf. **3**, 33 (2011).
[48]B. J. McBride, *NASA Glenn Coefficients for Calculating Thermodynamic Properties of Individual Species* (National Aeronautics and Space Administration, 2002).
[49]A. L. Kuhl and B. Khasainov, "Quadractic model of thermodynamic states in SDF explosions," Report No. UCRL-CONF-231294 (Lawrence Livermore National Laboratory, Livermore, CA, 2007).
[50]J. F. Baytos, "Specific heat and thermal conductivity of explosives, mixtures, and plastic-bonded explosives determined experimentally," Report No. LA-8034-MS (Los Alamos Scientific Laboratory, NM, 1979).
[51]N. Bergan, P. Butler, and H. Dwyer, "CHEMKIN real-gas: A FORTRAN package for analysis of thermodynamics and chemical kinetics in high-pressure systems," Report No. SAND91-8634 (Sandia National Laboratory, 1991).
[52]B. E. Poling, J. M. Prausnitz, and J. P. O'Connell, *Properties of Gases and Liquids* (McGraw-Hill Education, 2001).
[53]D. Chen, K. Wang, and H. Wang, "Violation of collision limit in recently published reaction models," Combust. Flame **186**, 208–210 (2017).
[54]B. Thornber, A. Mosedale, D. Drikakis, D. Youngs, and R. J. Williams, "An improved reconstruction method for compressible flows with low Mach number features," J. Comput. Phys. **227**, 4873–4894 (2008).
[55]Z. Shen, W. Yan, and G. Yuan, "A robust HLLC-type Riemann solver for strong shock," J. Comput. Phys. **309**, 185–206 (2016).
[56]R. J. Spiteri and S. J. Ruuth, "A new class of optimal high-order strong-stability-preserving time discretization methods," SIAM J. Numer. Anal. **40**, 469–491 (2002).
[57]P. N. Brown, G. D. Byrne, and A. C. Hindmarsh, "VODE: A variable-coefficient ODE solver," SIAM J. Sci. Stat. Comput. **10**, 1038–1051 (1989).
[58]W. Zhang, A. Almgren, V. Beckner, J. Bell, J. Blaschke, C. Chan, M. Day, B. Friesen, K. Gott, D. Graves *et al.*, "AMReX: A framework for block-structured adaptive mesh refinement," J. Open Source Software **4**, 1370–1370 (2019).
[59]R. W. Houim and K. K. Kuo, "A low-dissipation and time-accurate method for compressible multi-component flow with variable specific heat ratios," J. Comput. Phys. **230**, 8527–8553 (2011).
[60]R. W. Houim and E. S. Oran, "A multiphase model for compressible granular–gaseous flows: Formulation and initial tests," J. Fluid Mech. **789**, 166–220 (2016).
[61]J. D. Koch, S. Piecuch, J. M. Lightstone, J. R. Carney, and J. Hooper, "Time-resolved measurements of near infrared emission spectra from explosions: Pure pentaerythritol tetranitrate and its mixtures containing silver and aluminum particles," J. Appl. Phys. **108**, 036101 (2010).
[62]A. Kuhl, R. Ferguson, and A. Oppenheim, "Gasdynamic model of turbulent exothermic fields in explosions," Adv. Combust. Sci. **173**, 251–261 (1997).
[63]F. Ree, W. Pitz, M. Van Thiel, and P. Souers, "Overabundance of carbon monoxide in calorimetry tests," J. Phys. Chem. **100**, 5761–5765 (1996).

30 September 2026 21:40:58

Published under an exclusive license by AIP Publishing